PromptScaffold: Helping Non-Expert Users Write LLM Prompts with Structured Templates

Mira Castellan, Joon-ho Reyes and Priya Valcourt
School of Computing, Halden University of Technology

Abstract. People without training in machine learning increasingly write prompts for large language model (LLM) tools, but often struggle to state the task, give examples and specify the output format. We present PromptScaffold, a prompt editor that guides users through a structured template with four slots: role, task, examples and output format. In a within-subjects lab study with 16 participants who completed three predefined tasks, prompts written with PromptScaffold reached a higher automatic task score than free-form prompts (mean 0.74 vs. 0.61) and participants reported higher confidence. We discuss what the template helped with, where it constrained participants, and the limits of a single short lab study.

1 Introduction

LLM-based tools let anyone describe a task in natural language, yet writing an effective prompt remains difficult for people without machine-learning experience. Prior studies report that non-expert users tend to give instructions that are too short, rarely include examples, and leave the output format implicit. They then struggle to tell whether a poor result comes from the prompt or from the model.

Structured templates are a common answer in practice, but there is little evidence on whether they help non-experts write better prompts or only make prompts longer. This paper asks two questions. RQ1: Do prompts written with a structured template perform better on predefined tasks than free-form prompts? RQ2: How do non-expert users experience writing prompts with a template?

We contribute (1) PromptScaffold, a template-based prompt editor; (2) a within-subjects lab study with 16 participants; and (3) observations on where templates help and where they constrain users.

2 Related Work

Work on end-user programming shows that novices benefit from scaffolds that make implicit requirements explicit, such as forms and worked examples. Studies of prompt writing describe non-experts as relying on trial and error and on conversational phrasing. Prompt libraries and prompt-pattern catalogues offer reusable structures, but they are usually written for developers and have rarely been evaluated with non-expert users in a controlled setting.

3 The PromptScaffold Editor

PromptScaffold splits a prompt into four slots. The *role* slot states who the model should act as. The *task* slot describes what to do, with a hint to name the input and the goal. The *examples* slot asks for one to three input-output pairs, and the *output format* slot asks how the answer should be structured. Each slot shows a short hint and an example. The editor assembles the slots into one prompt that the user can preview and edit freely before running it. A side panel shows the model's output for a small set of test inputs that the user provides.

4 Study

Participants. We recruited 16 participants (9 women, 7 men, aged 19 to 27) through university mailing lists. All were undergraduate or master's students; 11 studied computing or engineering and 5 studied other subjects. None had taken a course on machine learning, and all had used a chatbot before.

Design and tasks. The study used a within-subjects design with two conditions: free-form prompting in a plain text box, and PromptScaffold. Each participant completed three predefined tasks in each condition: classifying customer reviews by sentiment, labelling support tickets by topic, and summarising short news articles in two sentences. Condition order was counterbalanced, and parallel task versions were used so that no participant saw the same inputs twice. All prompts were run on one LLM with fixed settings.

Procedure and measures. Each session lasted about 70 minutes in our lab. The main outcome was an automatic task score between 0 and 1: accuracy against gold labels for the two classification tasks and a reference-based overlap score for summaries. After each condition, participants rated their confidence in their prompts on a 7-point scale. We also recorded the time to the final prompt and held a short interview at the end of the session.

5 Results

Prompts written with PromptScaffold reached a higher task score than free-form prompts (Table 1). The difference was largest for the topic-labelling task, where the examples slot led most participants to add labelled examples. Participants needed slightly more time with the template. Mean confidence was 4.1 with PromptScaffold and 3.4 with free-form prompting.

Measure	Free-form	PromptScaffold
Task score, sentiment (0-1)	0.72	0.80
Task score, topic labels (0-1)	0.55	0.73
Task score, summaries (0-1)	0.56	0.69
Mean task score (0-1)	0.61	0.74
Confidence (1-7)	3.4	4.1
Time to final prompt (min)	6.2	7.5

Table 1: Mean results per condition (16 participants, three tasks each).

In the interviews, 12 participants said the examples slot was the most useful part of the template. Six said the role slot felt artificial, and four said the template made them write long prompts even for simple tasks. Several participants copied the hint text into the slots without adapting it.

6 Discussion

The template mainly helped by prompting participants to give examples and to state the output format, which free-form prompts often lacked. It did not teach participants why these parts mattered: several filled the slots mechanically. Designers of prompt tools for non-experts may therefore want to explain the purpose of each slot and let users drop slots that do not fit the task.

7 Limitations

Our findings have three main limitations. First, the sample was small and homogeneous: 16 university students, most of them from computing courses, so the results may not hold for older users or for people outside a university. Second, each participant used PromptScaffold in a single 70-minute lab session; we did not study long-term or real-world use, and novelty may have raised confidence. Third, we used only three predefined classification and summarisation tasks and one LLM, and measured success only with an automatic task score and self-reported confidence. Open-ended tasks, other models and expert ratings of prompt quality may give different results.

8 Conclusion

PromptScaffold's structured template improved task scores and confidence for non-expert users on three predefined tasks in a short lab study. Future work should test the template with more diverse users, over longer periods and on open-ended tasks, and should examine how to explain the purpose of each slot.